Research Goals and Interests. My research focuses on developing robotics manipulation systems that are practical, general, and robust. I am a Master's Student at the **University of Washington** co-advised by **Prof. Dieter Fox** and **Prof. Ranjay Krishna**. Notably during this period, I originated the idea, proposed, and led the development of **MolmoAct**[1] – from its inception to its release. Through MolmoAct, I became involved with the **Allen Institute for AI**, first as a Student Collaborator, and later as a part-time Student Researcher. I'm fortunate to have the opportunity to collaborate with **Prof. Florian Shkurti** while he was a Visiting Research Scientist at AI2.

In recent years, robot learning and large-scale data-driven methods have shown remarkable progress in manipulation. Through MolmoAct, I've experienced firsthand how scaled-up policies trained via imitation learning can perform dexterous manipulation tasks, execute subtle recovery maneuvers, and capture the richness and complexity of the real world. At the same time, I have also experienced the limitations of relying solely on data scaling. Real-world data at scale is expensive and difficult to collect, often accessible only to well-funded industry labs. While simulation data is easier to generate, it suffers from the sim-to-real gap and demands substantial environment design and engineering effort. Furthermore, unlike animal and human intelligence, which has evolved over millions of years to interact with the world through rich, high-dimensional sensing and rapid motor capabilities, current models lack a comparable level of embodied intelligence embedded in space and time. Towards that end, my research direction is driven by the following questions:

1. How can we build and leverage scalable data-driven methods for robotics manipulation?
2. How can we ground robotics foundation models with the rich priors and inductive biases already available from physics and geometry, together with more hierarchical and structured forms of reasoning and model architecture?
3. How can we integrate high-dimensional perception — from vision, to fingertip tactile feedback, to the compliance and forces at the wrist — into scalable data and model architectures?

I aim to explore how these diverse approaches can **complement** one another and how we might develop **hybrid manipulation systems** that combine the strengths of both **model-free** and **model-based** methods.

My Research. Inspired by one of his talks at CVPR 2024, I reached out to Prof. Ranjay Krishna and began working with him before starting my Master's program. Early on, I investigated how different modalities – such as code[3], 2D points[5], and 2D lines[2] – could serve as action space representation for robot policies. MolmoAct grew out of these ideas. It first learns 2D pointing and 2D bounding box prediction as auxiliary tasks during pretraining to understand spatial embodied concepts in images. In addition, MolmoAct leverages depth estimation and 2D visual traces as intermediate reasoning representations, allowing the robot to "think before acting."

For a large-scale, data-driven project like MolmoAct, designing streamlined and distributed infrastructure was essential. I built and designed MolmoAct's procedural data generation pipeline from the ground up and curated the MolmoAct Pretraining Dataset. To support experimentation and rapid iteration, I independently developed MolmoAct's parallelized simulation infrastructure for both pretraining and post-training, and led all real-world and simulation experiments. Throughout the project, I explored various modes and combinations of intermediate reasoning representations, as well as data mixtures and sampling rates. This work also taught me how to rigorously inspect large datasets by building infrastructure that allows me to inspect data quickly. For example, I discovered that while the Open-X-Embodiment[4] provides a useful foundation, it suffers from several quality issues—including low-resolution images, misannotated language instructions, excessive no-op actions, and toy environments.

To better align MolmoAct with real household tasks, we collected the MolmoAct Dataset –10k trajectories of single-arm robots across diverse household environments. I led the effort to design the set of household tasks; each task was grounded in an actual household chore and decomposed into subtasks (e.g., for "clean up all the dishes after dinner": put the bowl in the dishwasher, put the fork in the sink, cover the pot), enabling the policy to learn each component effectively. We also incorporate multimodal action distributions and recovery behaviors to capture diverse ways of accomplishing the same tasks. Ultimately, MolmoAct achieves substantial gains in policy performance by grounding actions in spatial reasoning, enabling more precise and steerable control.

Towards the end of MolmoAct, it became clear that while the model often understood what needed

to be done, it lacked the precise low-level control skills required to execute tasks reliably. Moreover, MolmoAct's architecture was not designed for the rapid execution speeds needed for fine-grained motor control, and existing robotics foundation models often lacked the polished, systematized scaling laws and training stages seen in frontier AI models. This realization made me increasingly fascinated by several unanswered questions: How do we build a reliable and scalable data-generation pipeline for robot demonstrations? How can we systematically study the scaling properties and effects of different data mixtures and sampling strategies, an area that currently feels more like art than science? How do we acquire contact-rich and dexterous manipulation skills, especially non-prehensile behaviors that require fine-grained, low-level control. And what model architectures are best suited to leverage large-scale robot data? These questions led me to my next project with Prof. Dieter Fox, where we began investigating these challenges directly.

We developed a procedural data-generation pipeline in IsaacSim that integrates TAMP with low-level RL policies. I trained state-based RL skills that focus on contact-rich, non-prehensile behaviors – such as pushing, flipping, and nudging – skills that are difficult for traditional motion planners, which struggle to model contact dynamics. During data generation, TAMP calls RL skills in place of motion planning whenever contact-rich skills are required. Our goal is to distill these demonstrations into visuomotor policies using Visual(RGB)-based RL, behavior cloning and even VLA models, and to evaluate their semantic, control, generalization capabilities on long-horizon manipulation tasks.

Moreover, I am leading the effort to investigate how force-torque sensing and compliance can be incorporated into both the data-generation pipeline and the resulting distilled models. Specifically, I am interested in several questions: 1) To what extent does force-torque sensing improve contact-rich, low-level skills, and how can it help in partially observable scenarios? 2) How can we increase robot execution speed, and does force-torque sensing enable reliable control at higher velocities? 3) Does force-torque sensing lead to more robust sim-to-real transfer, and can policies equipped with force-torque inputs generalize across variation in physical properties such as mass, friction and object shape? Taken together, these efforts reflect my broader goal of incorporating high-dimensional perception into scalable data and model architectures for manipulation systems, while grounding these models in stronger physical priors. Force–torque sensing at scale exposes models to contact dynamics that vision alone cannot capture, potentially leading to more robust policies.

[1] **Jason Lee** et al. *MolmoAct: Action Reasoning Models that can Reason in Space*. 2025. arXiv: 2508.07917 [cs.RO].

[2] Yi Li et al. "HAMSTER: Hierarchical Action Models for Open-World Robot Manipulation". In: *The Thirteenth International Conference on Learning Representations*.

[3] Jacky Liang et al. "Code as Policies: Language Model Programs for Embodied Control". In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2023, pp. 9493–9500.

[4] Abby O'Neill et al. "Open X-Embodiment: Robotic Learning Datasets and RT-X Models: Open X-Embodiment Collaboration". In: *ICRA*. 2024.

[5] Wentao Yuan et al. "RoboPoint: A Vision-Language Model for Spatial Affordance Prediction in Robotics". In: *Conference on Robot Learning*. PMLR. 2025, pp. 4005–4020.